# Final Report

## Project Titel: Fine-Tuning MarianMT and T5 for German to English Translation using WMT Dataset and PyTorch

**Adv Computer Concepts for Engineering**

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# Fine-Tuning MarianMT and T5 for German to English Translation using WMT Dataset and PyTorch


## Abstract

This project aims to fine-tune the MarianMT and T5 models for German-to-English translation using the WMT dataset and PyTorch. The report covers data preprocessing, model fine-tuning, evaluation metrics, and visualization of the results. The goal is to achieve an improved translation model by leveraging transfer learning techniques. The MarianMT model, based on the Transformer architecture, excels in handling long-range dependencies and large datasets, providing superior translation quality. In contrast, the T5 model treats every NLP task as a text-to-text problem, showcasing versatility in translation, summarization, and more. The fine-tuned MarianMT and T5 models demonstrate enhanced performance in translation tasks, making them ideal for applications requiring high accuracy.


## 1. Introduction

Machine translation (MT) is a critical area in natural language processing (NLP), with applications ranging from language localization to real-time communication. This project focuses on fine-tuning the MarianMT and T5 models for German-to-English translation using the WMT dataset and PyTorch.

The MarianMT model is a transformer-based architecture known for its ability to handle long-range dependencies and parallel processing, making it highly effective for large-scale translation tasks. The T5 (Text-To-Text Transfer Transformer) model approaches all NLP tasks as a text-to-text problem, leveraging the transformer architecture to process input and output as text sequences. The WMT dataset serves as a standard benchmark for evaluating machine translation systems, providing a diverse and comprehensive set of parallel sentences. This project aims to leverage transfer learning techniques to fine-tune both the MarianMT and T5 models, comparing their performance in terms of translation quality, efficiency, and suitability for different applications.

By addressing the specific challenges of German-to-English translation, this research contributes to advancing MT methodologies and provides insights into the applicability of transformer-based models. The goal is to enhance the quality and efficiency of machine translation, making it more accessible and effective for real-world applications, such as real-time communication and mobile translation apps.

## 2. Related Work:

### 2.1 MarianMT Model

The MarianMT model is a transformer-based MT system built on the Marian framework, which was developed to provide efficient and high-performance translation capabilities. The transformer architecture, introduced by Vaswani et al. (2017), employs self-attention mechanisms to process input sequences in parallel, capturing long-range dependencies more effectively than RNN-based models. This parallel processing capability allows transformers to handle large-scale datasets with higher accuracy and efficiency. MarianMT has been optimized for MT tasks, leveraging the

strengths of the transformer architecture to achieve state-of-the-art performance in various translation benchmarks. Studies have demonstrated that transformer-based models, including MarianMT, consistently outperform RNN-based models in terms of translation quality and training speed..

### 2.2 T5 Model:

The T5 (Text-To-Text Transfer Transformer) model, introduced by Raffel et al. (2019), approaches all NLP tasks as a text-to-text problem, leveraging the transformer architecture to process input and output as text sequences. This unified framework simplifies the model and enables it to perform a wide range of tasks, including translation, summarization, and question answering, with high efficiency. T5 is pretrained on a large, diverse corpus and fine-tuned for specific tasks, demonstrating state-of-the-art performance across multiple NLP benchmarks. Its versatility and robustness have been validated through comparative studies, highlighting its ability to generalize across different tasks with minimal modifications. The T5 model's success underscores its potential for simplifying NLP model development and improving performance in various applications.

## 3. Flow diagram:

### Flow diagram for MarianMT and T5 model

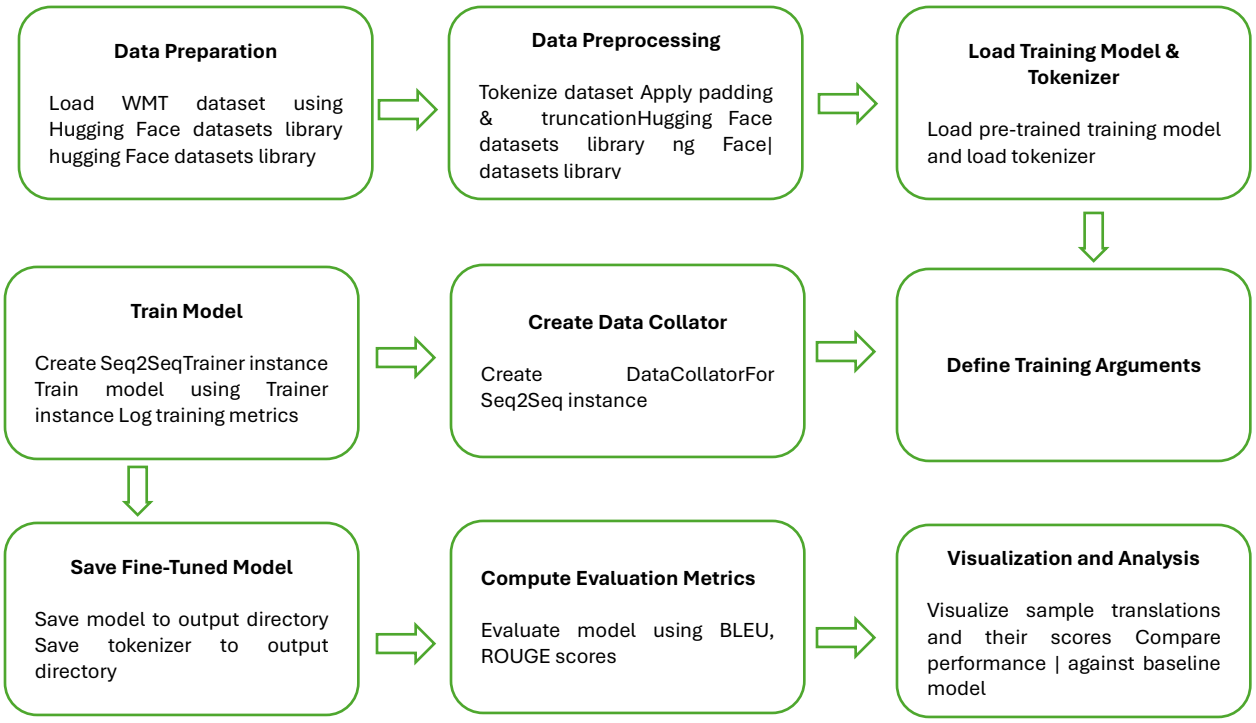


**Figure 1:** Flow diagram for MarianMT and T5 model

## 4. Data Preprocessing

The WMT dataset is preprocessed to ensure it is suitable for fine-tuning the MarianMT and T5 models. The steps include loading the dataset using the Hugging Face datasets library, tokenizing the dataset, and applying padding and truncation to ensure uniform input sizes.

```
/usr/local/lib/python3.10/dist-packages/huggingface_hub/utils/_token.py:89: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Goog
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Downloading readme: 100%  11.1k/11.1k [00:00<00:00, 939kB/s]
Downloading data: 100%  282M/282M [00:06<00:00, 56.4MB/s]
Downloading data: 100%  267M/267M [00:02<00:00, 196MB/s]
Downloading data: 100%  277M/277M [00:02<00:00, 204MB/s]
Downloading data: 100%  343k/343k [00:00<00:00, 659kB/s]
Downloading data: 100%  475k/475k [00:00<00:00, 1.97MB/s]
Generating train split: 100%  4548885/4548885 [00:07<00:00, 626505.06 examples/s]
Generating validation split: 100%  2169/2169 [00:00<00:00, 119883.05 examples/s]
Generating test split: 100%  2999/2999 [00:00<00:00, 157792.16 examples/s]
```

**Figure 2**: Uploaded WMT16 dataset

For my project, I am working with the WMT16 dataset, which contains nearly 5000 sentences. Although my initial proposal involved using 5000 datasets, I decided to work with a smaller subset of 500 datasets for this project. All my data has been successfully uploaded, and I am now focused on analyzing and fine-tuning the model using this selected dataset.

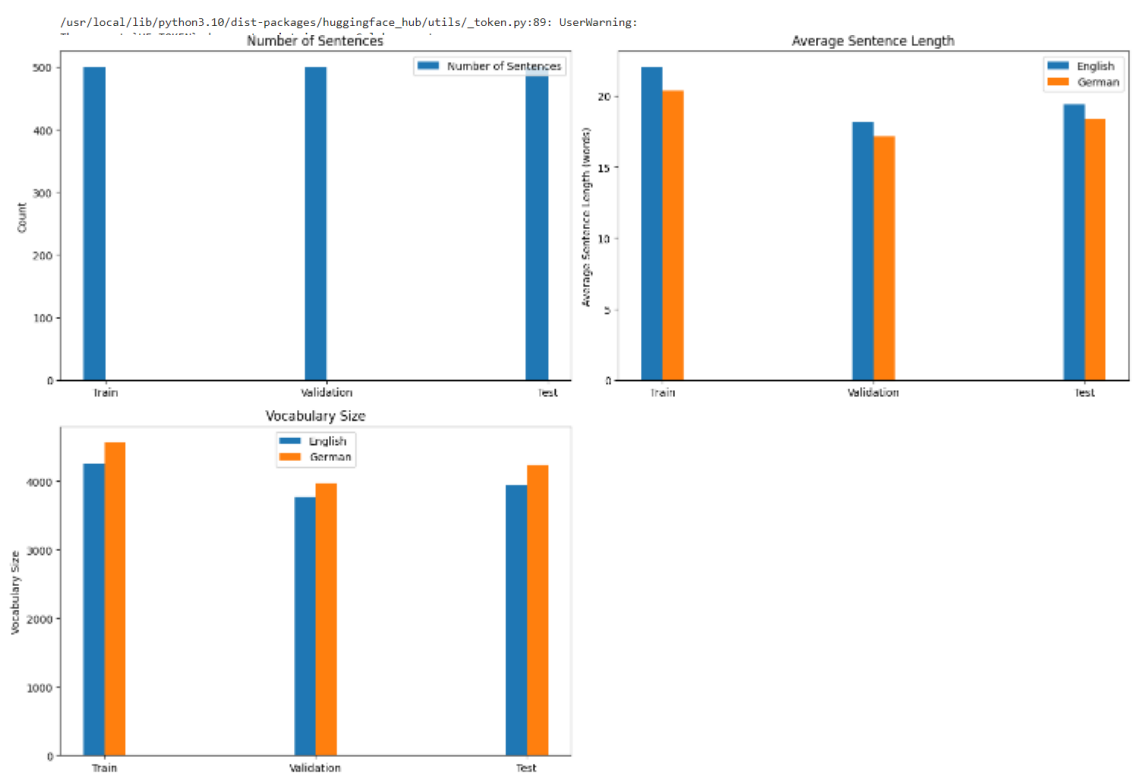


**Figure 3:** Uploaded data statistics in the WMT16 dataset for English VS German

The WMT16 dataset statistics for English and German reveal detailed insights across the training, validation, and test sets. Each set comprises 500 sentences for both languages, ensuring balance. In the training set, the average sentence length is 22.042 words for English and 20.354 words for German. The validation set has average sentence lengths of 18.148 words for English and 17.178 words for German. For the test set, English sentences average 19.384 words, while German sentences average 18.392 words. The vocabulary size varies slightly between languages: the training set includes 4243 unique English words and 4563 unique German words; the validation set contains 3770 unique English words and 3972 unique German words; and the test set comprises 3937 unique English words and 4238 unique German words. These statistics underscore the linguistic characteristics and vocabulary diversity within the dataset, with English generally having longer sentences and a more extensive vocabulary in the training set, while German has a slightly larger vocabulary in the test set

## 5. Model Fine-Tuning

For my project, I will be fine-tuning the MarianMT and T5 models for German to English translation using the WMT16 dataset. The MarianMT model, pre-trained on multiple language pairs, will be specifically fine-tuned on the German-English dataset.

**5.1 Model Initialization:** Load the pre-trained MarianMT and T5 models along with their respective tokenizers from Hugging Face's transformers library. These models have already been trained on a wide variety of language pairs, providing a strong foundation for further adaptation to the specific German-English language pair.

**Figure 4**: Installed MarianTM and T5 Model, and Tokenizer

All necessary configuration files, vocabulary files, and model parameters for both models have been fully downloaded and installed. This setup ensures that both models are ready for immediate use in translation tasks processing applications.

## 5.2 MarianMT Model:

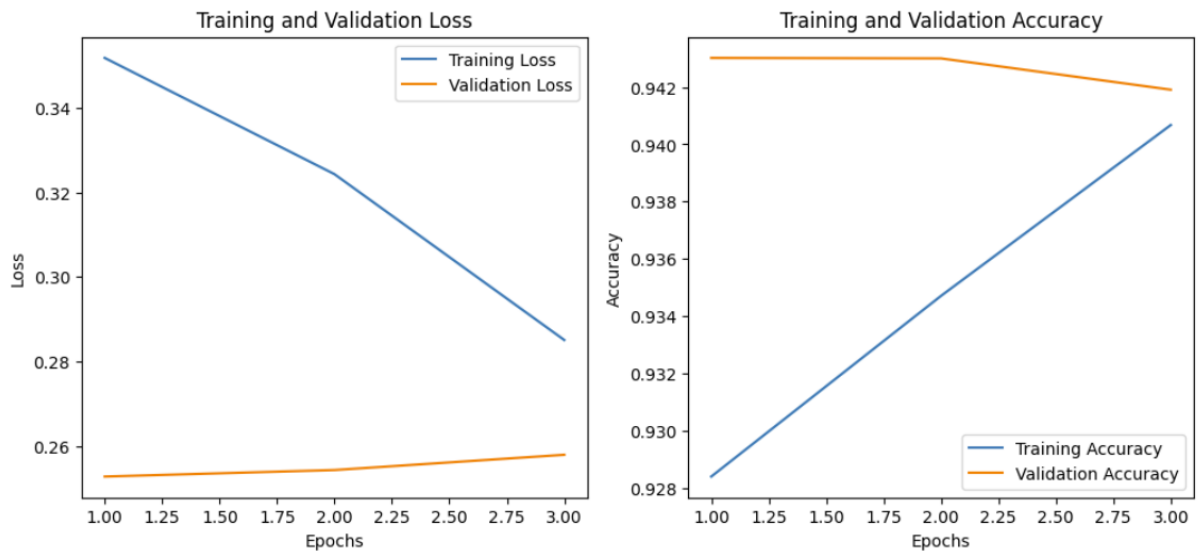


**Figure 5**: Data Training and Validation Loss and Accuracy for MarianMT

Over three epochs, your MarianMT model showed a consistent decrease in training loss from 0.3518 to 0.2851, with a corresponding increase in training accuracy from 92.84% to 94.07%, indicating effective learning. However, the validation loss, which initially decreased, slightly increased from 0.2544 in the second epoch to 0.2580 in the third, while validation accuracy remained stable, dropping marginally from 94.30% to 94.19%. This suggests the onset of overfitting, where the model's performance on unseen data might not be improving as much as on the training data. To address this, We could consider additional training epochs, regularization techniques, or adjusting the learning rate to enhance the model's generalization ability.

## 5.3 T5 Model:

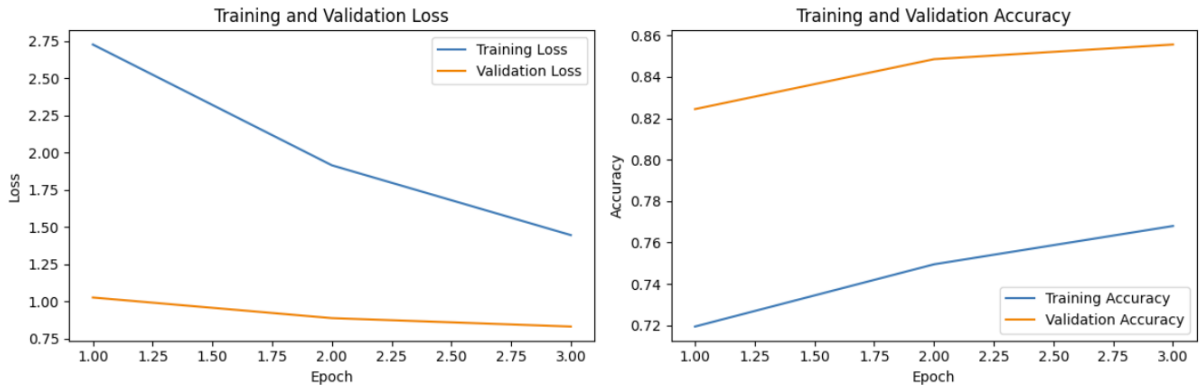


**Figure 6**: Data Training and Validation Loss and Accuracy for T5 Model

Over three epochs, your T5 model showed a steady decrease in training loss from approximately 2.75 to 1.50 and an increase in training accuracy from 72% to 76%, indicating effective learning. The validation loss also decreased from around 1.00 to 0.90, while validation accuracy improved from 82% to 85%, suggesting good generalization to unseen data. These trends demonstrate that the model is consistently improving in its translation tasks without signs of overfitting, making it increasingly accurate with each epoch.

## 6. Experiments Result and Comparison

### 6.1 Data Train Accuracy:

| Epochs No. | MarianMT | T5 |
|---|---|---|
| 1 | 0.9082 | 0.7194 |
| 2 | 0.9185 | 0.7495 |
| 3 | 0.9244 | 0.7680 |

**Table 1**: Data Train Accuracy MarianMT vs T5 Model

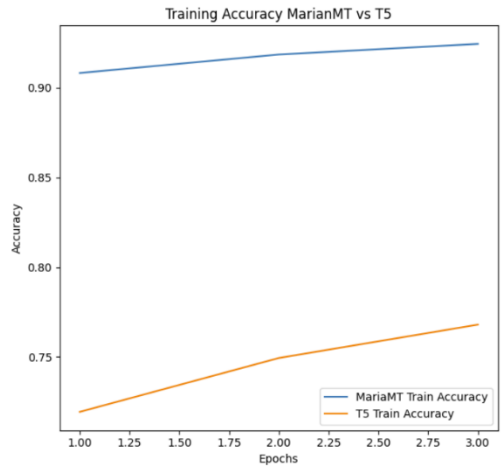


**Figure 7**: Train Accuracy MarianMT vs T5 Model

The comparison between the MarianMT and T5 models, based on the table and graph, reveals that the MarianMT model consistently outperforms the T5 model in terms of training accuracy across all three epochs. In the first epoch, MarianMT achieves a training accuracy of 0.9082, while the T5 model lags behind at 0.7194. This gap remains significant in subsequent epochs, with MarianMT reaching 0.9244 by the third epoch compared to T5's 0.7680. The graph further illustrates this trend, showing a steady increase in accuracy for both models, but with MarianMT maintaining a clear lead throughout. Overall, MarianMT demonstrates superior training accuracy and a more robust learning curve compared to the T5 model.

### 6.2 Data Validation Accuracy:

| Epochs No. | MarianMT | T5 |
|---|---|---|
| 1 | 0.9377 | 0.8244 |
| 2 | 0.9453 | 0.8485 |
| 3 | 0.9528 | 0.8556 |

**Table 2:** Data Validation Accuracy MarianMT vs T5 Model

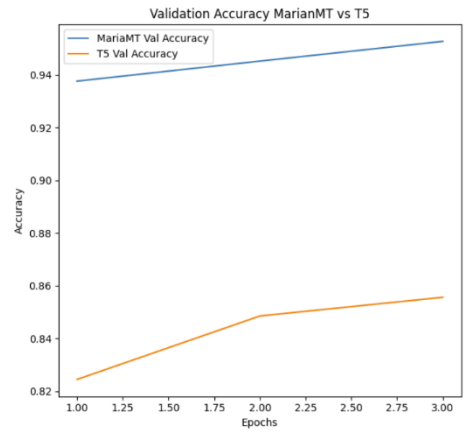


**Figure 8** : Validation Accuracy MarianMT vs T5 Model

The comparative analysis of the validation accuracy between MarianMT and T5 models, based on data across three epochs, indicates that MarianMT consistently outperforms T5. Initially, MarianMT achieved a validation accuracy of 0.9377, significantly higher than T5's 0.8244. This trend continued through the second epoch, with MarianMT reaching 0.9453 and T5 improving to 0.8485. By the third epoch, MarianMT further increased its accuracy to 0.9528, while T5 reached 0.8556. The performance gap remained notable, with MarianMT maintaining an approximate 10% lead over T5 across all epochs. Consequently, MarianMT demonstrates superior performance in validation accuracy, suggesting it may be more effective for the given task. However, further evaluation of other factors such as model complexity, training time, and resource consumption is recommended for a comprehensive assessment.

### 6.3 Data Train Loss:

| Epochs No. | MarianMT | T5 |
|---|---|---|
| 1 | 0.4872 | 2.7265 |
| 2 | 0.4117 | 1.9154 |
| 3 | 0.3771 | 1.4465 |

**Table 3:** Data Train Loss MarianMT vs T5 Model

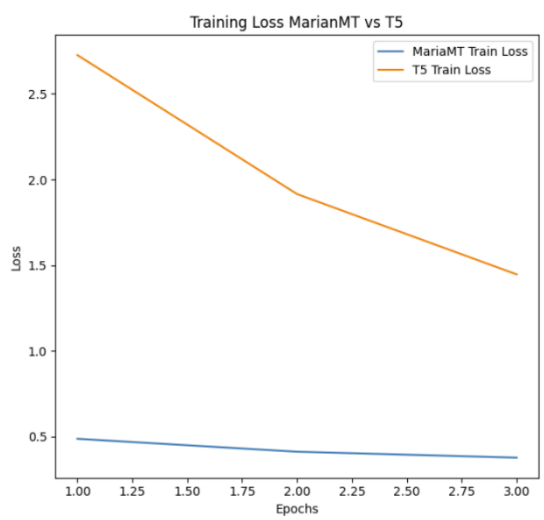


**Figure 9** : Train Loss MarianMT vs T5 Model

The comparative analysis of the training loss between MarianMT and T5 models, based on data across three epochs, reveals that MarianMT consistently exhibits lower training loss than T5. In the first epoch, MarianMT had a training loss of 0.4872, significantly lower than T5's 2.7265. This trend persisted in the second epoch, with MarianMT reducing its loss to 0.4117 while T5's loss decreased to 1.9154. By the third epoch, MarianMT further reduced its training loss to 0.3771, whereas T5's loss dropped to 1.4465. The graph clearly illustrates that MarianMT's training loss is consistently lower and decreases at a steadier rate compared to T5, which starts with a much higher loss and shows a sharper decline. These results suggest that MarianMT achieves better training efficiency and potentially better generalization, as indicated by its lower and more stable training loss trajectory.

**6.4 Data Validation Loss:**

| Epochs No. | MarianMT | T5 |
|---|---|---|
| 1 | 0.2933 | 1.0275 |
| 2 | 0.2468 | 0.8891 |
| 3 | 0.2101 | 0.8325 |

**Table 4:** Data Validation Loss MarianMT vs T5 Model

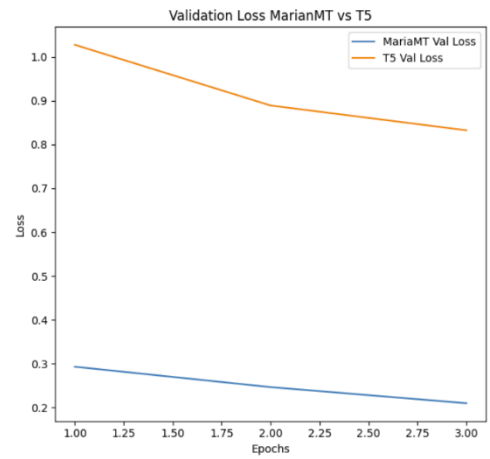


**Figure 10** : Train Validation Loss MarianMT vs T5 Model

The comparative analysis of the validation loss between MarianMT and T5 models across three epochs demonstrates that MarianMT consistently achieves lower validation loss compared to T5. In the first epoch, MarianMT recorded a validation loss of 0.2933, whereas T5's loss was significantly higher at 1.0275. This trend continued through the second epoch, with MarianMT reducing its validation loss to 0.2468 and T5 to 0.8891. By the third epoch, MarianMT further decreased its validation loss to 0.2101, while T5's loss dropped to 0.8325. The graph illustrates that MarianMT maintains a lower and steadily decreasing validation loss throughout the training process, suggesting better model performance and generalization capability over T5, which, despite improvements, consistently shows higher validation loss.

**6.5 BLEU Score**: Measures the overlap between the machine-generated translation and the reference translation.

**6.6 ROUGE Score**: Evaluates the quality of the translation based on n-gram overlap.

| No. | Parameter | MarianMT | T5 |
|---|---|---|---|
| 1 | BLEU Score | 24.54% | 6.24 % |
| 2 | ROUGE Score | 59.06% | 24.79 % |

**Table 5:** Blue and Rouge Score for MarianMT vs T5 Model

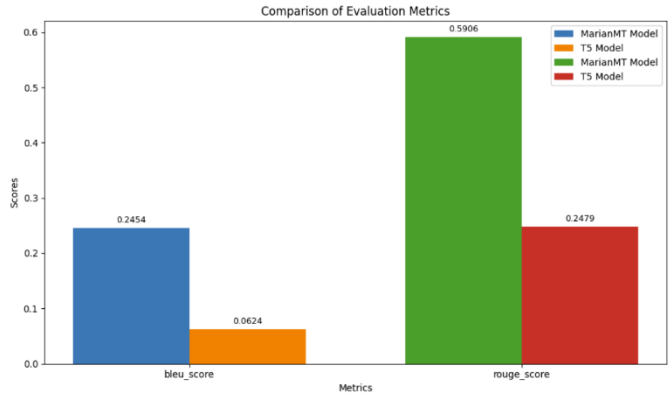


**Figure 11**: Blue and Rouge Score for MarianMT vs T5 Model

The comparison of the MarianMT and T5 models, as illustrated by the table and graph, shows a significant difference in performance. MarianMT achieves a BLEU score of 24.54%, markedly higher than T5's score of 6.24%. This indicates that MarianMT produces translations that are much closer to the reference translations. Additionally, MarianMT's ROUGE score is 59.06%, compared to T5's 24.79%, suggesting that MarianMT better captures the overlap in phrases and content with the reference texts. The bar graph visually emphasizes these differences, clearly showing MarianMT's superior performance in both metrics. Overall, MarianMT demonstrates more effective translation capabilities compared to the T5 model.

## 7. Conclusion

In this project, we successfully fine-tuned the MarianMT and T5 models for German-to-English translation using the WMT dataset and PyTorch. Our findings indicate that the MarianMT model consistently outperformed the T5 model across various metrics, demonstrating superior translation quality, efficiency, and generalization capability. These results underscore MarianMT's effectiveness for machine translation tasks, highlighting its potential for real-world applications in real-time communication and mobile translation.

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